Integrated Claims AI Platform: Multi-Service Integration Architecture

Five service domains converge on the unified claims record in Azure AI Search. Every arrow is labelled with the data it carries.

data flow · event flow · audit / compliance logs · source · storage · Azure AI service · destination

1 · DOCUMENT INTELLIGENCE

Claims documents
Portal & e-mail uploads, mailroom scans

PDF · TIFF · JPEG claim packets

Blob landing zone
raw-intake/{claimId} · immutable originals · malware scan · CMK

blob URL (Event Grid trigger)

Document Intelligence: Custom Classifier
splitMode = auto · classes: invoice, claims form, police report, medical record, other

page ranges + doc type + confidence

Document Intelligence: Extraction
prebuilt-invoice + query fields · claims forms: composed template models · police & medical: custom neural · low confidence → LLM fallback / human review

typed fields JSON + field confidence

2 · VISION

Damage photos
Mobile app & portal: vehicle and property claims

JPEG / HEIC images + EXIF

Photo store (Blob)
claim-photos/{claimId} · annotated copies for adjuster UI

image SAS URL

Azure AI Vision: Image Analysis 4.0
captions · tags · objects · Read OCR (VIN, plates) · multimodal embeddings

image vectors + OCR (duplicate check)

captions, tags, OCR text

Azure OpenAI: GPT-4o-class vision
severity reasoning over full photo set + claim context · JSON-schema output · advisory only

severity grade + rationale + damaged parts JSON

image SAS URL

Custom damage detector
Azure ML AutoML object detection (dent, crack, hail, water, fire...)

damage boxes + class + confidence

3 · SPEECH

Claims intake calls
Contact-centre telephony · en / es / fr / pt

live audio stream (8 kHz PCM)

Azure AI Speech: real-time STT
Custom Speech per locale · continuous language ID · diarization

live partial transcript

Agent-assist desktop
live transcript · FNOL form pre-fill

call recording (WAV)

final diarized transcript

Call-audio store (Blob)
CMK · private endpoint · WORM · retained 6 yrs after claim closure

audio SAS URL (batch job)

Azure AI Speech: batch transcription
same custom models · diarization · output to own storage (BYOS)

diarized transcript JSON

Azure AI Language: post-call analytics
PII redaction · sentiment · NER · key phrases · conversation summary

redacted transcript + call insights JSON

Historical call archive
Recorded calls (WAV)

legacy WAV files

4 · NLP

Customer communications
E-mail · web chat · SMS · letters: en / es / fr / pt

message text + channel metadata

Azure AI Language
language detection · PII redaction · sentiment & opinion mining · NER · custom text classification (e-mail) · CLU (chat / IVR)

redacted source text + language code

Azure AI Translator
inbound → English · Custom Translator glossary · Document Translation · outbound EN → es / fr / pt

message JSON: intents, entities, sentiment, original + EN text

audio access & disposal logs

translated replies (es / fr / pt)

5 · KNOWLEDGE MINING

Policy & regulatory archive
200,000 documents: PDF, DOCX, scanned TIFF

PDF / DOCX / TIFF + metadata

kb-archive (Blob)
4 virtual-folder partitions · soft delete · version metadata

new & changed blobs

Azure AI Search indexers (×4)
batchSize 20 · LastModified change detection · soft-delete detection · enrichment cache

cracked text + page images

Enrichment skillset
Built-in: Document Layout, Entity Recognition, Key Phrases, Azure OpenAI Embedding · Custom: document classification, claims metadata

chunks + vectors + metadata (index projections)

enriched shapes: tables, objects, files

Azure Service Bus claim-events topic (subscriptions: claims system, analytics)

claim events JSON (review flags, severity, intents, complaints)

routing & review events

INTEGRATION LAYER Azure Functions (Durable) · normalise to claims-record schema · push API · publish events

document artifact JSON (push API, mergeOrUpload)

photo artifact JSON

call artifact JSON

communication artifact JSON

AZURE AI SEARCH
private endpoint · Entra RBAC (API keys off)

★ CONVERGENCE POINT: UNIFIED CLAIMS RECORD

claims-record index

key artifactId · **join key** claimId

artifactType document | photo | call | communication

Document Intelligence: claimNumber, policyNumber, dateOfLoss, lossType, invoiceTotal, vendorName, reportNumber, diagnosisCodes

Vision: damageClasses, severityGrade, totalLossFlag, captions, duplicatePhotoScore, imageVector

Speech: redactedTranscript, callSummary, callerSentiment, entities, language

NLP: intents, sentiment, originalText, englishText

Security fields (filterable, not retrievable): assignedAdjusterIds, adjusterGroups, sensitivity

Retrieval: BM25 + vector (RRF) · semantic ranker · vectorFilterMode = preFilter

Producers: integration layer pushes each artifact as it is processed (mergeOrUpload, claimId-keyed)

Consumers: Claims API (adjuster workbench) · claim context for GPT severity prompt

policy-regulatory-kb index
≈4M clause chunks · HNSW vectors (int8) · scoring profile *adjusterRelevance* · semantic config · security-filter fields

query + security filter ⇄ ranked, trimmed results

user access audit (who, claim, documents)

index & query logs

Claims Management System
work queues · adjuster assignment · complaint register · SIU referrals

Claims API (APIM + App Service)
Entra ID auth · resolves assigned claims → mandatory security filter · issues 15-min audio SAS

query + Entra token ⇄ results, captions, audio SAS link

Claims Adjuster Search Interface
claims workbench: unified claim view, policy & regulation search, audio playback

event stream: sentiment, intents, call metrics

Analytics & Reporting
Microsoft Fabric · Power BI: call sentiment, intent trends, cycle time, clause usage

projection tables (Power BI)

Knowledge Store (Azure Storage)
tables: Documents, Clauses, KeyPhrases, Entities · objects: enriched JSON · files: page images

COMPLIANCE LOGGING Azure Monitor Log Analytics workspace ← diagnostic settings on every resource (Storage, Document Intelligence, Vision, Speech, Language, Translator, Azure OpenAI, Search, Key Vault) + application access audit · Microsoft Sentinel alerts (bulk audio download, unassigned-claim access, deletes outside disposal job) · immutable log archive, retained 6 years (23 NYCRR 500.6)

SECURITY & GOVERNANCE PLANE Microsoft Entra ID (RBAC, PIM, MFA, Conditional Access) · Key Vault customer-managed keys · private endpoints / VNet integration · Azure Policy (deny public access & shared-key auth) · all data in US regions